



Punjab University College
of
Information Technology
“PUCIT”



Chapter 15 – Computer Security and Privacy

GE-161 Introduction to Information and Communication Technologies

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Overview

- This chapter covers:
 - Hardware loss, hardware damage, and system failure, and the safeguards that can help reduce the risk of a problem occurring due to these concerns
 - Software piracy and digital counterfeiting and steps that are being taken to prevent these computer crimes
 - Possible risks for personal privacy violations due to databases, marketing activities, electronic surveillance, and monitoring, and precautions that can be taken to safeguard one's privacy

Why Be Concerned About Computer Security?

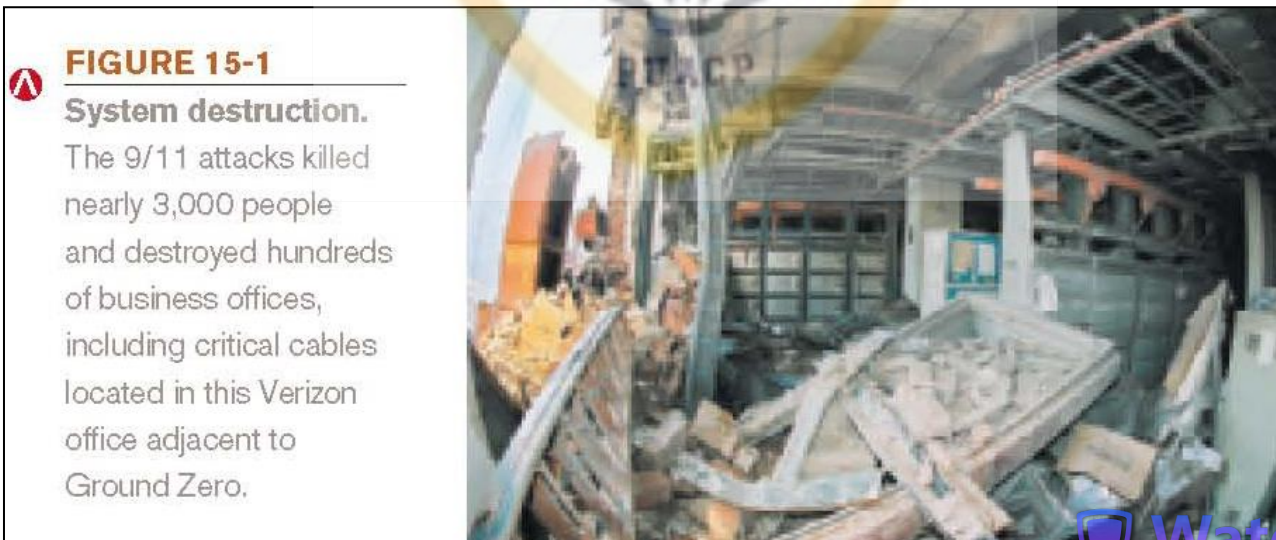
- There are a number of security concerns related to computers that users should be aware of, including:
 - Having a computer or other device stolen
 - Losing important data
 - Losing contact lists
 - Buying pirated or counterfeited products

Hardware Loss, Damage, and System Failure

- Hardware loss: Can occur when a personal computer, USB flash drive, mobile device, or other piece of hardware is stolen, lost, or damaged
 - Hardware theft: When hardware is stolen from an individual or an organization
 - Often stolen from homes, businesses, cars, airports, hotels, etc.
 - Often occurs for the value of the hardware, but increasingly for the information that might be contained on the hardware

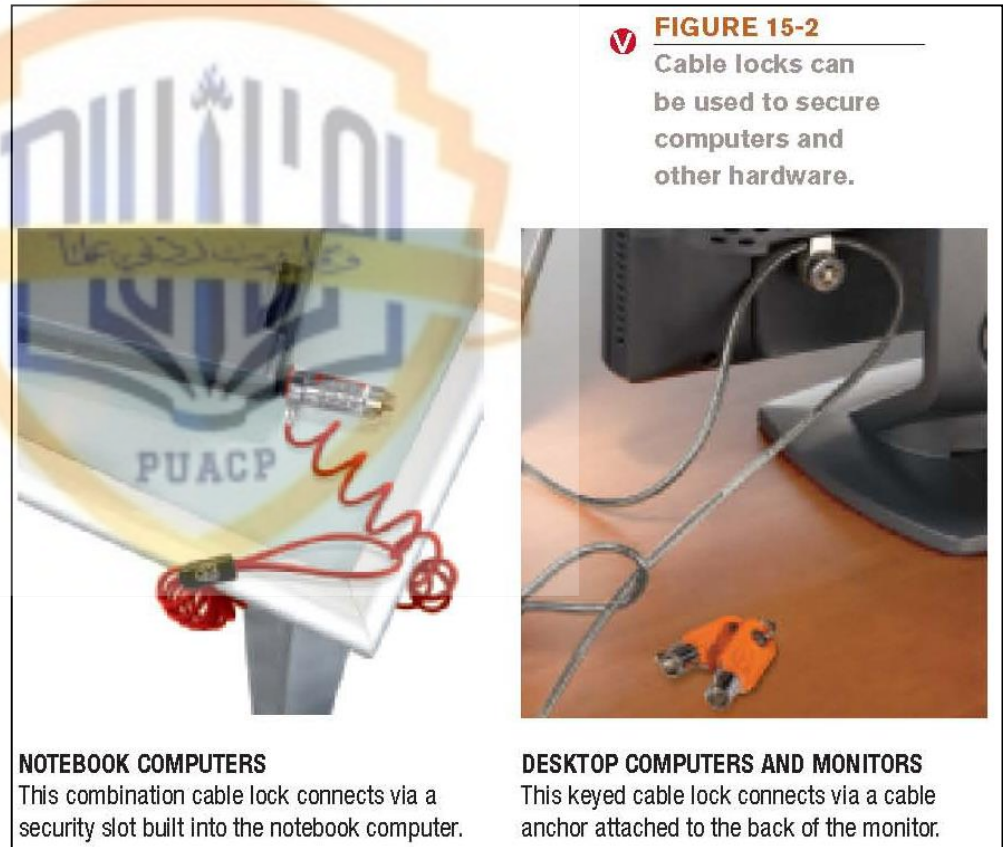
Hardware Loss, Damage, and System Failure

- Hardware damage: Can be accidental or intentional
- System failure: The complete malfunction of a computer system
 - Can be due to a hardware problem, software problem, or computer virus
 - Can be due to a natural disaster or planned attack



Protecting Against Hardware Loss, Hardware Damage, and System Failure

- Use door and computer equipment locks
 - Cable locks
 - Security slots
 - Cable anchors
 - Laptop alarm software
 - Lock up USB flash drives, external hard drives, and other media



Protecting Against Hardware Loss, Hardware Damage, and System Failure

- Use encryption to protect data
 - Increasingly used with USB flash drives, portable computers, external hard drives, etc.
 - Full disk encryption (FDE): Everything on the storage medium is encrypted
 - Self-encrypting hard drive: A hard drive using FDE



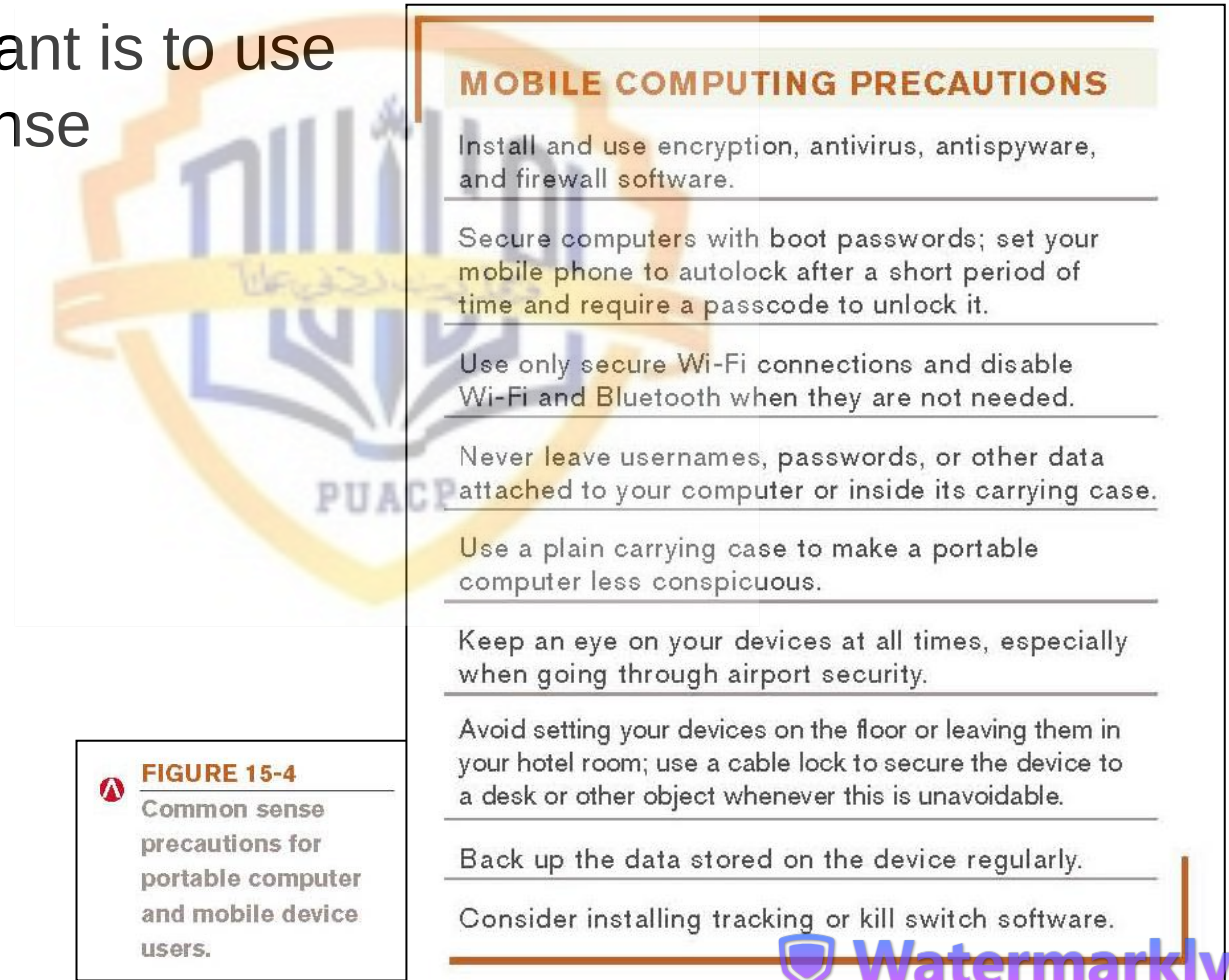
FIGURE 15-3
Encrypted media.
The data on this encrypted USB flash drive cannot be accessed until the user is authenticated via a fingerprint scan.

Protecting Against Hardware Loss, Hardware Damage, and System Failure

- Computer tracking software: Used to find a computer or other device after it is lost or stolen
 - Sends out identifying data via the Internet
 - Law enforcement can use this data to recover the device
 - Stealth tracking software: The sending of data is transparent to the user
 - Kill switch: Technology that causes the device to self-destruct
- Other precautions:
 - Asset tags, tamper evident labels, etc.

Protecting Against Hardware Loss, Hardware Damage, and System Failure

- Additional precautions for mobile users
 - Most important is to use common sense



Protecting Against Hardware Loss, Hardware Damage, and System Failure

FIGURE 15-5
Protective cases.

- Proper hardware care
 - Don't abuse hardware
 - Use protective cases
 - Ruggedized devices available



MOBILE PHONE CASES



NOTEBOOK CASES



RUGGED PORTABLE COMPUTERS



RUGGED MOBILE DEVICES



RUGGED MOBILE PHONES

FIGURE 15-6
Ruggedized devices.

Protecting Against Hardware Loss, Hardware Damage, and System Failure

- Surge suppressors:
Protect hardware from damage due to electrical fluctuations
- Uninterruptible power supplies (UPSs):
Provide continuous power to a computer system for a period of time after the power goes off



Protecting Against Hardware Loss, Hardware Damage, and System Failure

– Also:

- Watch dust, moisture, static, heat, etc.
- Avoid head crash
- Stop USB devices before removing
- Use screen protectors, jewel cases, etc.

FIGURE 15-8
Proper hardware care. Unless your computer is ruggedized (such as the one shown here), keep it out of the heat, cold, rain, water, and other adverse conditions.



Protecting Against Hardware Loss, Hardware Damage, and System Failure

- Backup and disaster recovery plans:
 - Both businesses and individuals should use appropriate backup procedures
 - Backup media needs to be secured
 - Data storage companies store backup media at secure remote locations
 - Online backup is another possibility
 - Continuous data protection (CDP): Enables data backups to be made on a continual basis
 - Disaster-recovery plan: Describes the steps a company will take following the occurrence of a disaster

Quick Quiz

1. Which of the following would not likely be a reason for stealing a notebook computer?
 - a. For the data contained on the computer
 - b. To use in a denial of service (DoS) attack
 - c. For the value of the hardware
2. True or False: It is only important to use a surge suppressor during bad weather, when a lightning strike may occur.
3. A copy of a file that is created in case the original is damaged is called a(n) _____.

Answers:

1) b; 2) False; 3) backup

Software Piracy and Digital Counterfeiting

- Software piracy: Unauthorized copying of a computer program
 - Occurs when:
 - Individuals make illegal copies of software to give to friends
 - Businesses or individuals install software on more than the number of computers allowed according to the end-user license agreement (EULA)
 - Sellers install unlicensed copies on computers sold to consumers
 - Large-scale operations in which programs and packaging are illegally duplicated and sold as supposedly legitimate products

Software Piracy and Digital Counterfeiting

- Digital counterfeiting: The use of computers or other types of digital equipment to make illegal copies documents
 - Currency, checks, collectibles and other items
 - Often scanned and printed or color-copied

FIGURE 15-10
Digital counterfeiting.
Documents commonly counterfeited include currency, credit cards, driver's licenses, passports, and checks.



Protection Against Software Piracy and Digital Counterfeiting

- Software antipiracy tools
 - Educating businesses and consumers
 - Registration code or product key
 - Checking validity of a software installation before upgrades or other resources related to the program can be used
 - Watching online auction sites/lawsuits
 - Incorporating code into applications to inform the vendor when pirated copies are being used, or is in violation of the license

Protection Against Software Piracy and Digital Counterfeiting



FIGURE 15-11

Antipiracy software.

This software identifies businesses (including their physical locations) using a pirated version of the vendor's software.

Protection Against Software Piracy and Digital Counterfeiting

- Digital counterfeiting prevention
 - New currency designs
 - Microprinting, watermarks, security thread, etc.
 - Special paper is used with U.S. currency
 - Identifying technology included in digital imaging hardware
 - Digital watermarks: Subtle alteration to a digital item that is not noticeable but that can be retrieved to identify the owner of the item
 - Also can use:
 - RFID tags and other hard-to-reproduce content

Protection Against Software Piracy and Digital Counterfeiting

MICROPRINTING

Extremely small print that is very difficult to reproduce appears in three different locations on the front of the bill (in the left and right borders, at the top of the shield on the Great Seal, and in between the columns of the shield), though it is hard to see without a magnifying glass.

SECURITY THREAD

A plastic security thread embedded in the paper contains the letters "USA" followed by the number "5"; it can be seen when the bill is held up to the light and glows blue when placed in front of an ultraviolet light.



COLORS

Harder to match colors, such as shades of yellow and purple, have been added to some details.

WATERMARKS

Watermark images containing the number "5" located to the right and left of the portrait are incorporated into the paper itself and are visible when the bill is held up to the light (not visible in this photograph).

FIGURE 15-12
Anticounterfeiting measures used with U.S. currency.

Quick Quiz

1. Using a computer to make illegal copies of currency to circulate as real currency is a form of _____.
 - a. software piracy
 - b. computer sabotage
 - c. digital counterfeiting
2. True or False: Software piracy is rarely performed today.
3. Printed text or an image on a CD case or box that changes its appearance when viewed from different angles to prove the item is authentic is called a(n) _____.

Answers:

1) c; 2) False; 3) hologram

Why Be Concerned About Information Privacy?

- Privacy: State of being concealed or free from unauthorized intrusion
- Information privacy: Rights of individuals and companies to control how information about them is collected and used
- Computers add additional privacy challenges
 - Many data breaches recently due to lost or stolen hardware, carelessness with documents containing sensitive data, etc.
 - Businesses need to be concerned with the expense, damage to reputation, and possible lawsuits

Databases, Electronic Profiling, Spam, and Other Marketing Activities

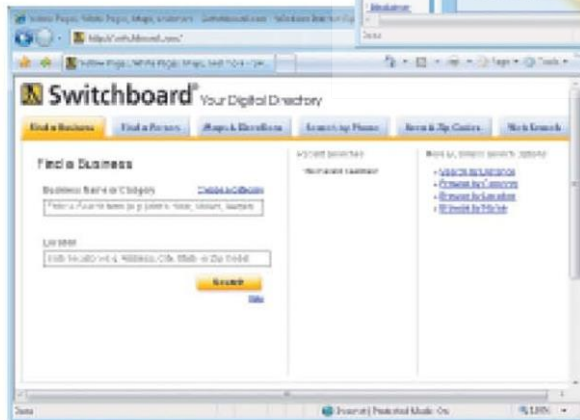
- Marketing database: Collection of data about people, used for marketing purposes
 - Data obtained through online and offline purchases, public information, etc.
 - Used in conjunction with Web activities
 - Social activity and searches performed
- Government database: Collection of data about people, collected and maintained by the government
 - Tax information, Social Security earnings, personal health records, marriage and divorce information
 - Some information is confidential, other is public

Databases, Electronic Profiling, Spam, and Other Marketing Activities



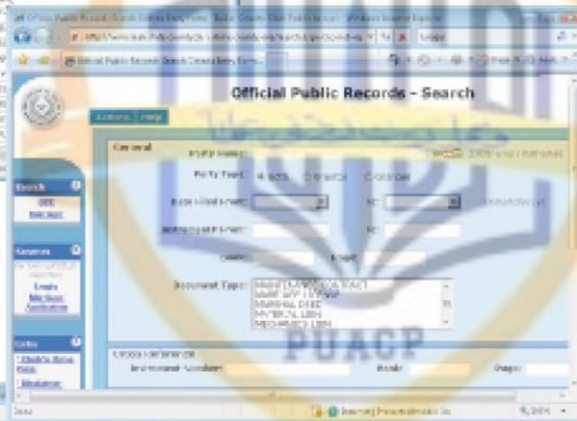
VITAL RECORDS SEARCH

Some counties and states allow searches for documents related to marriages, divorces, births, legal judgments, deeds, liens, powers of attorney, and so forth.



PROPERTY VALUE SEARCH

Some states permit searches for property located in that state, such as displaying the owner's name, address, and a link to additional information including property value for the supplied owner name.



ADDRESS NUMBER AND PHONE NUMBER SEARCH

Any information listed in a U.S. telephone book can be found using this site. You can search either by name or telephone number to view the available information.

FIGURE 15-13
A variety of searchable databases are available via the Internet.

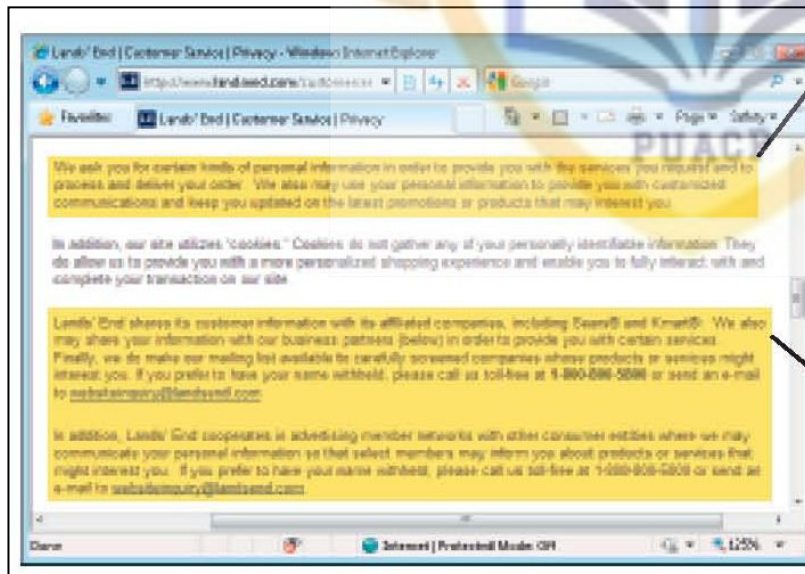
Databases, Electronic Profiling, Spam, and Other Marketing Activities

- Electronic profiling
 - Using electronic means to collect a variety of in-depth information about an individual



Databases, Electronic Profiling, Spam, and Other Marketing Activities

- Privacy policy: Discloses how information you provide will be used
 - Included on many Web sites
 - Dictates how supplied information may be used, but can be changed and often without notice



This indicates that your personal information may be used to keep you updated on products that might interest you.

This indicates that your personal information may be disclosed to third parties, unless you opt out.

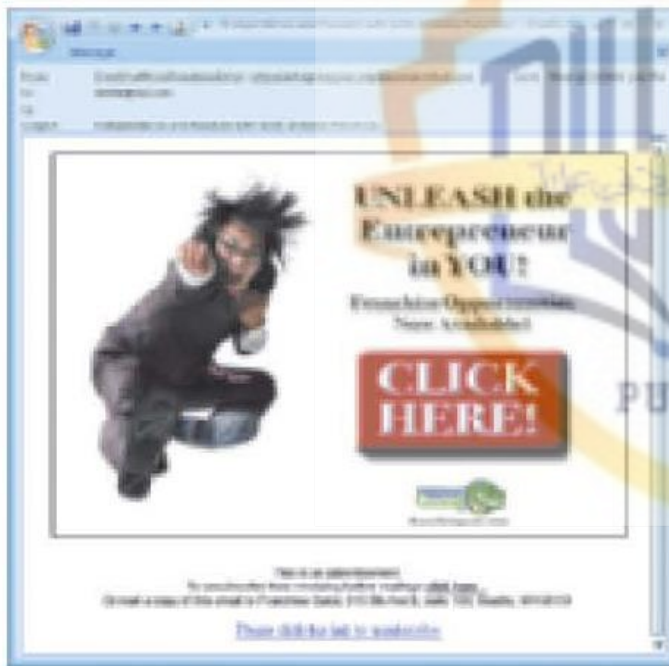
FIGURE 15-15
Privacy policies.
Web site privacy policies explain how your personal information might be used.

Spam and Other Marketing Activities

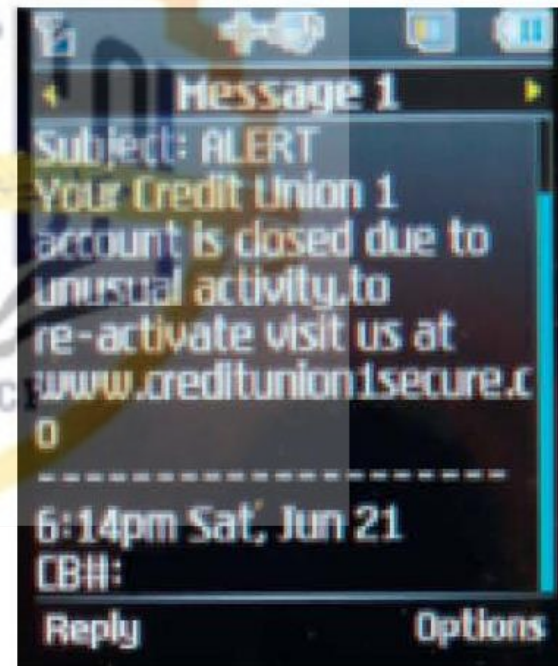
- Spam: Unsolicited, bulk e-mail sent over the Internet
 - Often involves health-related products, counterfeit products, fraudulent business opportunities, pornography, etc.
 - Marketing e-mails from companies a person has done business with
 - Appearing via instant messaging (spim)
 - Also delivered via mobile phones, social networking sites

Spam and Other Marketing Activities

✓ **FIGURE 15-16**
Examples of spam.



E-MAIL SPAM



TEXT MESSAGE SPAM

Protecting the Privacy of Personal Information

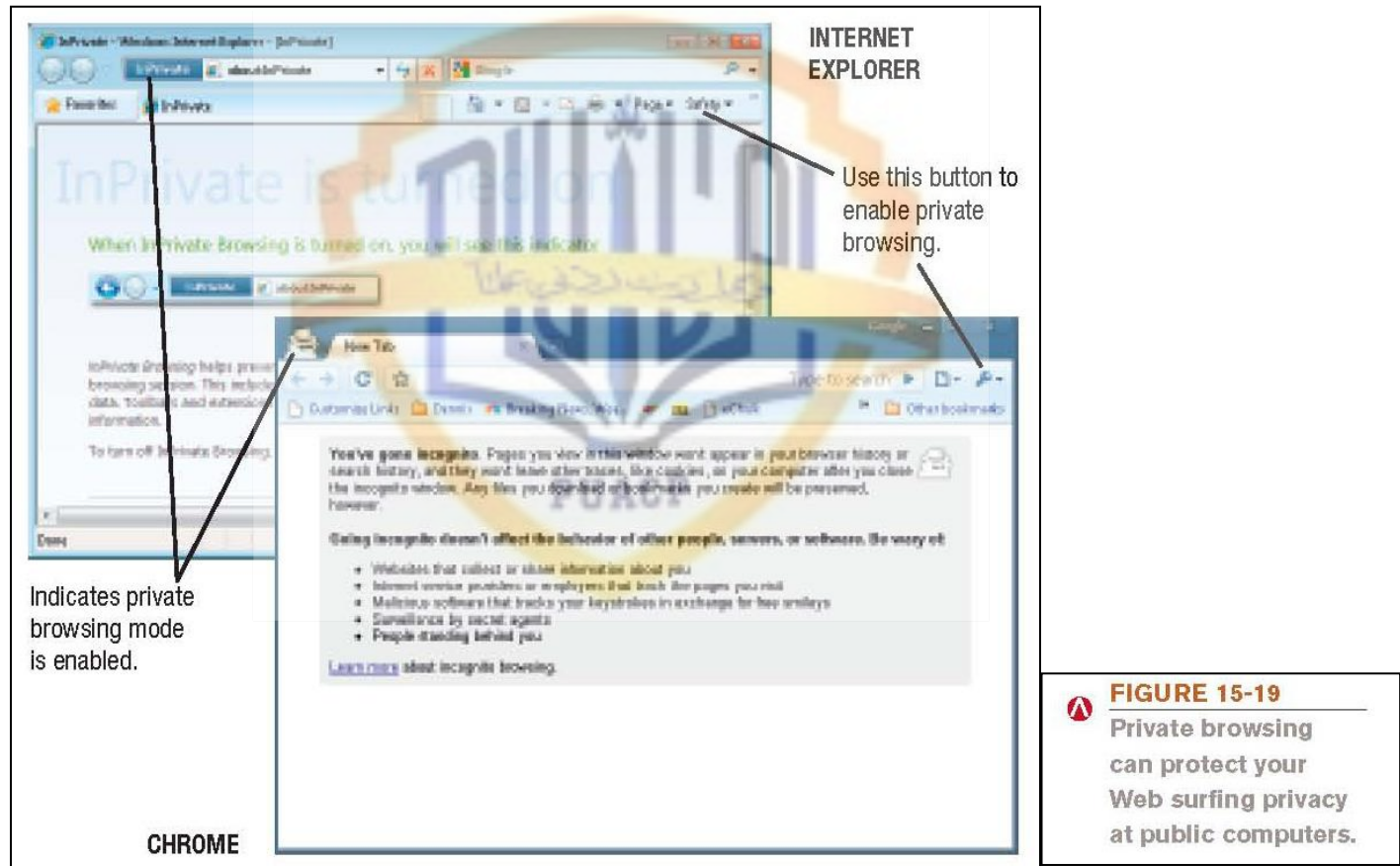
- Safeguard your e-mail address
 - Use a throw-away e-mail address (an extra e-mail address that you can use for activities that might result in spam)
 - Get a second e-mail address from your ISP or from Hotmail, Yahoo! Mail, or Gmail
 - Can stop using it and get a new one when needed



Protecting the Privacy of Personal Information

- Be cautious of revealing personal information
 - Read a Web site's privacy policy
 - Avoid putting too many personal details on your Web site
 - Be wary of sites offering prizes in exchange for personal information
 - Consider using privacy software, such as anonymous surfing
 - Supply only the required information in registration forms
 - Delete your browsing history and e-mail settings when using a public computer

Protecting the Privacy of Personal Information



Protecting the Privacy of Personal Information

- Use an e-mail filter to automatically route possible spam into a special folder to deal with later

The image shows a sequence of three overlapping dialog boxes from Microsoft Outlook, illustrating the process of creating an email filter. The first box, 'Rules Wizard', is in the foreground, showing 'Step 1: Select a template' with 'Many Organized' selected. The second box, 'Rules Wizard', is behind it, showing 'Step 2: Select conditions' with 'which contain specific words in the subject' selected. The third box, 'Rules and Alerts', is in the background, showing the 'Choose a folder' list with 'New Incoming (3,9)' selected. Arrows indicate the flow from the first box to the second, and then to the third.

1. Choose **Rules and Alerts** from the Tools menu, then click **New Rule** to start the Rules Wizard and create a new e-mail filter.

2. Select the desired conditions and actions, such as moving all messages containing certain terms into a specific folder.

3. The desired subject line terms and folder are selected here.

FIGURE 15-20
E-mail filtering with Microsoft Outlook.

Protecting the Privacy of Personal Information

- Can opt out from marketing activities
 - Some privacy groups want individuals to have to opt in to activities instead
- Web servers holding sensitive data should be secured
 - Only enter personal information on Web sites using secure servers
 - Automatic encryption systems for e-mail can help sensitive data from accidentally being revealed

Protecting the Privacy of Personal Information

- Properly dispose of hardware and outdated data
 - Wipe (not just delete) data on hard drives before disposing of a computer or hard drive
 - Storage media containing sensitive data should be shredded
 - Businesses should have a media sanitation/data destruction policy



FIGURE 15-21
Media disposal.

When disposing of CDs, DVDs, and other storage media, the media should be shredded to ensure the information on the media is destroyed.

Electronic Surveillance and Monitoring

- Computer monitoring software: Records an individual's computer usage either by capturing images of the screen, recording the actual keystrokes used, or creating a summary of Web sites visited
 - Can be used in homes by adults to monitor computer usage of children or spouse
 - Can be used in businesses to monitor employee computer usage
 - Keystroke-logging programs: Used to capture keystrokes
 - Can be used by hacker to capture usernames, passwords, and other sensitive information entered into a computer via the keyboard

Computer Monitoring Software

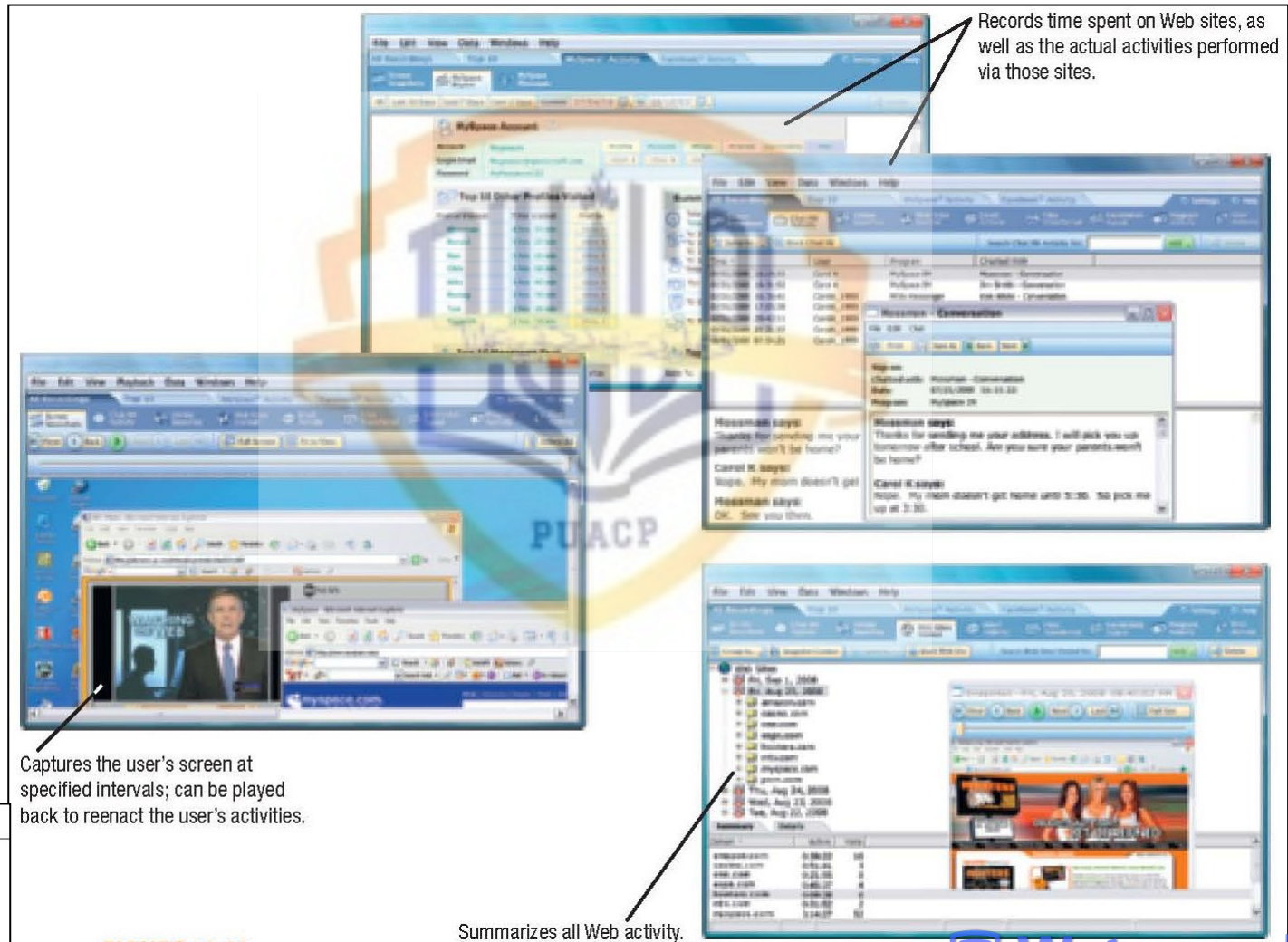


FIGURE 15-22
Computer monitoring software.

Electronic Surveillance and Monitoring

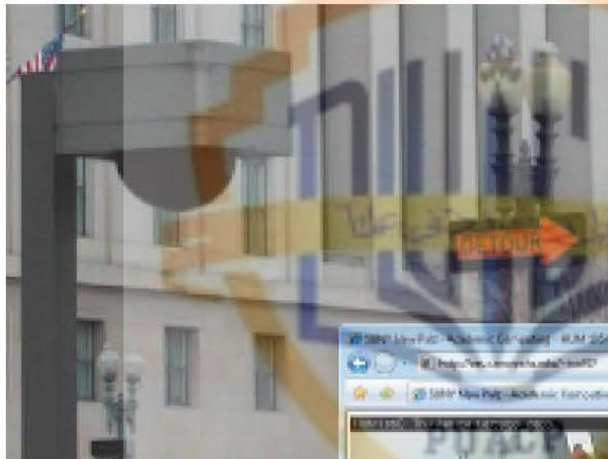
- Video surveillance: The use of video cameras to monitor activities of individuals
 - Used to monitor employees
 - Used in public locations for crime-prevention purposes
 - Stores and other businesses, Public streets
 - Subways, airports, etc.
 - Can be used with face recognition software
 - Identify terrorists and other known criminals
 - Privacy issues also involved with the use of camera phones

Video Surveillance



FIGURE 15-23

Examples of public video surveillance.

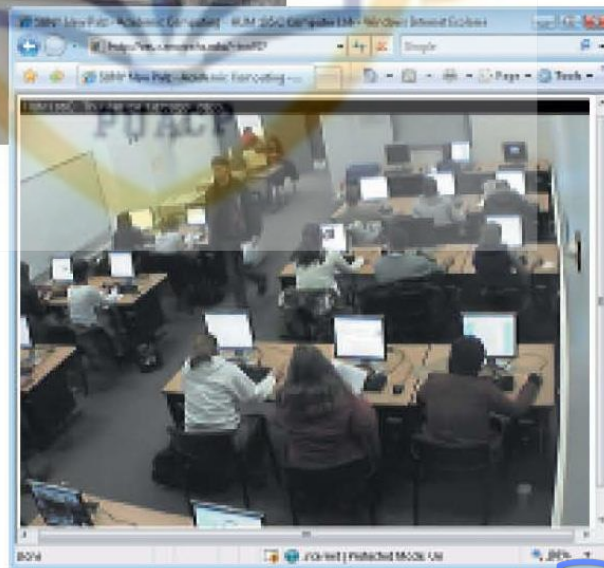


OUTDOOR SURVEILLANCE

Many cameras placed in public locations are designed to blend into their surroundings to be less intrusive, such as the camera inside this light fixture on a Washington, D.C., street.

INDOOR SURVEILLANCE

Many cameras are placed inside businesses, schools, and other locations, such as the ones that broadcast live video from several locations at a university in New York; a snapshot from one of the video cameras is shown here.



Electronic Surveillance and Monitoring

- Employee monitoring: Observing or recording employees' actions while they are on the job
 - Can monitor computer usage, phone calls, e-mail, etc.
 - Can monitor physical location
 - Video cameras
 - GPS capabilities built into cars or mobile phones
 - Proximity cards
 - Can also be used to access a facility, computer, etc.
 - Businesses should notify employees of monitoring



Electronic Surveillance and Monitoring

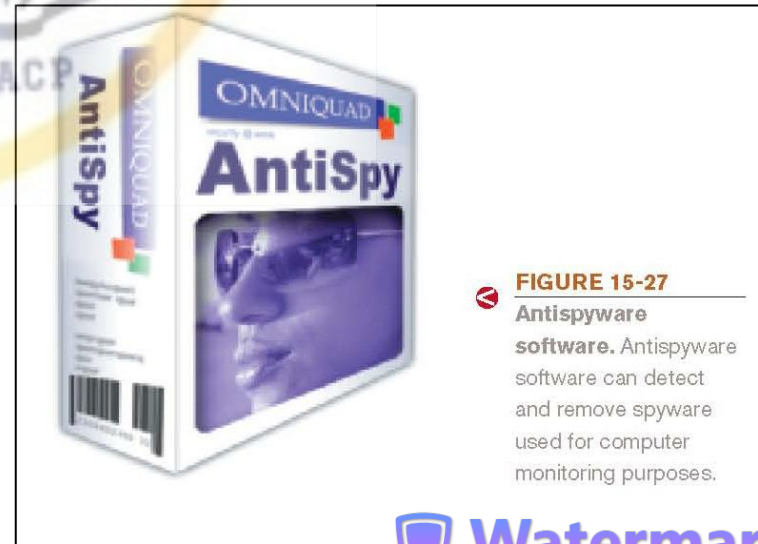
- Presence technology: Enables one computing device to locate and identify the **current status** of another device on the same network
 - Instant messaging, mobile phones, etc.
 - Can be used to **locate co-workers** or by customers
 - May also be used for **marketing activities** in the future
 - Potential privacy violations



FIGURE 15-26
Presence technology. Presence icons indicate the status of individual contacts.

Protecting Personal and Workspace Privacy

- Can use antispyware software to detect and remove some types of illegal computer monitoring and spyware software
- Employers have a responsibility to keep employee and customer information private and secured
- Employees should be familiar with their company's employee policy and avoid personal activities at work



Quick Quiz

1. A document that discloses how your personal information will be used is called a(n) _____.
 - a. privacy policy
 - b. opt out
 - c. throw-away e-mail address
2. True or False: The problem of protecting personal privacy and keeping personal information private did not exist before computers and the Internet.
3. The ability of one computing device on a network to identify the status of another device on that network is known as _____.

Answers:

1) a; 2) False; 3) presence technology

Summary

- Why Be Concerned About Computer Security
- Hardware Loss, Hardware Damage, and System Failure
- Software Piracy and Digital Counterfeiting
- Why Be Concerned About Information Privacy
- Databases, Electronic Profiling, Spam, and Other Marketing Activities
- Electronic Surveillance and Monitoring
- Computer Security and Privacy Legislation